A POWER TORSION MODULE IS UNCHANGED BY BASE CHANGE ALONG A MAP THAT IS AN ISOMORPHISM ON THE QUOTIENTS BY THE POWERS OF THE IDEAL

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 24172-24183 (canonical tag 0BNK) and the counted proof is the proof environment at source lines 24185-24198, which immediately follows it and is its own proof.

Editorial note. The counted statement contains the author's reference environment recording that the statement is a slight generalization of a result of Beauville and Laszlo; the retained preamble defines that environment as comment-type, so its body and the citation inside it are discarded when the document is typeset, which is why the card needs no bibliography.

Editorial note. The two references are to results of another chapter on commutation of tensor products with limits and on finitely generated ideals; the kinds recorded are the type words appearing in the labels themselves. Both are navigation stubs with their roles recorded in the reasons.

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Allowed standard prerequisites (editorial).

- A module annihilated by a power of the ideal is a module over the quotient, and tensoring it over the quotient against the base changed quotient gives back the module when the quotients agree. used at proof lines 24185-24190. Establishes the case of a module annihilated by a fixed power through the displayed chain of isomorphisms.
- A power torsion module is the union of the submodules annihilated by the successive powers of the ideal, and tensor products commute with direct limits. used at proof lines 24190-24196. Passes from the fixed powers to an arbitrary power torsion module.
- The completion of the ring with respect to a finitely generated ideal is a special case of the base change setting, namely the base change along the completion map. used at proof lines 24196-24198. Deduces the final assertion of the statement from the general one.

Source reference. Original lemma 00DD: sources/stacks-chow/tags.txt, source lines 493–493. The reference is invoked at proof lines 24190-24193 to pass from the

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modules annihilated by a fixed power to an arbitrary power torsion module; the chain of identifications for the fixed power is inside the counted span.

Source reference. Original lemma 05GG: sources/stacks-chow/tags.txt, source lines 7060–7060. The reference is cited at proof lines 24196-24198 for the final assertion, identifying the completion of a finitely generated ideal as an instance of the general setting; it is a pointer and the general statement is proved inside the counted span.

Lemma 0.2. *Let $R \to R'$ be a ring map. Let $I \subset R$ be an ideal such that $R/I^n \to R'/I^nR'$ is an isomorphism for $n > 0$. For any I-power torsion R-module M the map $M \to M \otimes_R R'$ is an isomorphism. For example, if I is finitely generated and $R^\wedge$ is the completion of R with respect to I, then we have $M \cong M \otimes_R R^\wedge$.*

Proof. If M is annihilated by I^n, then

$$M \otimes_R R' \cong M \otimes_{R/I^n} R'/I^nR' \cong M \otimes_{R/I^n} R/I^n \cong M.$$

If M is I-power torsion, then $M = \bigcup M[I^n]$. Since tensor products commute with direct limits (Algebra, Lemma 00DD), we obtain the desired isomorphism. The last statement is a special case of the first statement by Algebra, Lemma 05GG. □